

Data & Replication Models for Community-Led Climate Adaptation & Resilience

From citizen science to AI-driven early warning systems
— a practitioner's journey across geospatial analytics,
remote sensing & climate data engineering.

 Environmental

 Agriculture

 Climate Analytics

 Renewable Energy

Background, Training & Areas of Work



Foundation Degree

B.Tech — Geoinformation & Earth Observation Technology
Technical University of Kenya
Spatial analysis · Remote sensing · GIS · Environmental modelling



Advanced Training

Autonomous Agentic AI Systems

Multi-agent orchestration, tool-use, human-in-the-loop workflows

Data Science & Engineering

LuxDevHQ Academy · ETL/ELT pipelines · Cloud-native infrastructure · Streaming systems

CORE SKILL AREAS

Geospatial & Earth Observation

GEE · GeoPandas · Rasterio · Remote Sensing · STAC, GDAL

Data Engineering & Pipelines

Python · Dask · Airflow · PySpark · Kafka · dbt

AI/ML & Modelling

scikit-learn · PyTorch · SHAP · Ensemble models

Cloud & DevOps

AWS · GCP · Docker · CI/CD · Terraform · Monitoring

Agentic AI Systems

"My work sits at the intersection of Earth observation, AI engineering, and community empowerment."

Where It Began: Citizen Science & Urban Intelligence



Above-Ground Biomass Prediction

Nairobi, Kenya

- Predictive ML models using multi-spectral satellite imagery (Sentinel-2)
- Estimated carbon stock in urban & peri-urban trees across Nairobi
- NDVI, EVI, LAI as primary feature variables
- Random Forest Regressor — R^2 0.84 on independent validation set

EO data quantifies what ground surveys miss at scale



Community-Driven Street Tree Mapping

Citizen Science Platform

- Designed mobile-first citizen mapping platform for data collection
- Trained community volunteers as data collectors and validators
- Geocoded 12,000+ individual trees across Nairobi CBD
- Open dataset: species, canopy spread, health status, GPS coordinates

Communities as sensors — better coverage, local knowledge, data ownership



Urban Green Space Walkability

Mental Health & Spatial Equity

- Computed walkability scores to green spaces per neighbourhood
- Correlated with mental health survey data (anxiety, depression indices)
- Spatial regression—green accessibility a significant predictor ($p < 0.001$)
- Output: policy maps highlighting underserved districts

Data-driven urban planning directly improves citizen wellbeing

Multi-Indicator Remote Sensing & Decision Support Systems



MISLAND Africa DSS

Supported by GMES Africa

- Multi-indicator land degradation classification at continental scale
- Remote sensing indices: NDVI, NDWI, BSI, soil carbon trends
- Interactive DSS enabling non-technical policy users across Africa
- Direct architectural forerunner of the ARIN Climate DSS
- misland-africa.oss-online.org



Desert Locust Mapping & Modelling

Horn of Africa — Kenya · Somalia · Ethiopia

- Habitat suitability modelling using NDVI, rainfall & temperature
- MaxEnt + Random Forest ensemble — weekly swarm risk probability maps
- Delivered early warning layers to ICPAC/IGAD food security systems
- Climate-driven pest risk = climate-driven food security risk



Maternal Health Accessibility

Narok County, Kenya

- Network analysis: travel time to nearest maternal health facility
- Integrated road networks, elevation & seasonal flood risk layers
- Identified 38% of women with over 2 hours of travel to delivery services
- Informed county health facility placement strategy

3

54M

GMES

Community Data Ownership, Soil Intelligence & Energy



Community Soil Intelligence Platform

Amini · Sierra Leone

- Trained farming communities to collect geo-tagged soil fertility data
- Built ingestion & validation pipelines for community-sourced samples
- Interactive soil map viewer — farmers own their data layer
- Data-driven recommendations for fertiliser, crop selection & water use
- Improved data governance: consent, attribution, community benefit



Kenya Renewable Energy Pipeline

Solving data fragmentation in government energy planning
Unified pipeline: solar irradiance · wind atlas · grid infrastructure · land cover
odero54.github.io/kenya-energy-pipeline



Africa Geospatial Knowledge Graph

Building at Amini · Continental Soil Intelligence

- Semantic Knowledge Graph linking soil samples, EO data, crop outcomes & climate
- Enables AI agents to reason across heterogeneous African soil and climate data
- Foundation for AI-powered agronomic decision support at continental scale
- Addresses the core problem: data exists, but it is siloed and inaccessible

"Data governance is not a technical problem — it is a trust problem. Communities collect better data when they own it."

"The goal: every community in Africa with access to AI-powered, locally-owned climate intelligence."

Africa's Climate Crisis Demands a Data Revolution

46.3M

People affected by drought & flooding in Horn of Africa, 2020–2023

5

Consecutive failed rainy seasons — worst drought in 40 years (OCHA)

\$1.5B

Livestock loss in Kenya alone. 9.5 million animals dead across the region

20%

Of Africa's adaptation finance needs are currently being met (CPI 2024)

Fragmentation).

Climate data exists in disconnected silos. No unified access pathway for communities to query, combine, or act on it.

Latency

Early warnings often arrive after the crisis has begun. Communities need forecasts, not retrospectives.

Inaccessibility

Models are built for scientists. Farmers and pastoralists cannot interpret CDI anomalies—they need translated, actionable signals.

WHAT REPLICATION MODELS SOLVE

Open Data

ERA5, MODIS, Sentinel — free, global, 25+ year records — no licence fees

Modular Architecture

Each index (PDI, TDI, VDI) is independently replaceable as better data becomes available

Community Translation

Scientific CDI values → 5 plain-language severity classes → SMS/radio alerts

Forecast Capability

STL decomposition + SARIMA statistical model → 6-month drought outlook with confidence bounds

Composite Drought Index (CDI): A Replicable Multi-Index Framework

PDI

Precipitation Drought Index

ERA5-Land monthly precipitation anomaly vs. 25-year climatological baseline

TDI

Temperature Drought Index

ERA5-Land 2m temperature—captures heat stress component of compound drought

VDI

Vegetation Drought Index

MODIS MOD13A3 NDVI anomaly — vegetation stress response to rainfall and heat deficit

CDI

Composite Drought Index

Weighted combination of PDI, TDI, and VDI — the integrated multi-indicator drought signal

DATA PIPELINE — FROM SATELLITE TO COMMUNITY ALERT

1 Data Streaming

ERA5-Land (precip + temp) and MODIS NDVI streamed from Google Earth Engine — 25 years, monthly, global

2 Parallel Processing

Dask in-memory computation across all pixels simultaneously — no disk writes, fast spatial analysis at any scale

3 CDI Computation

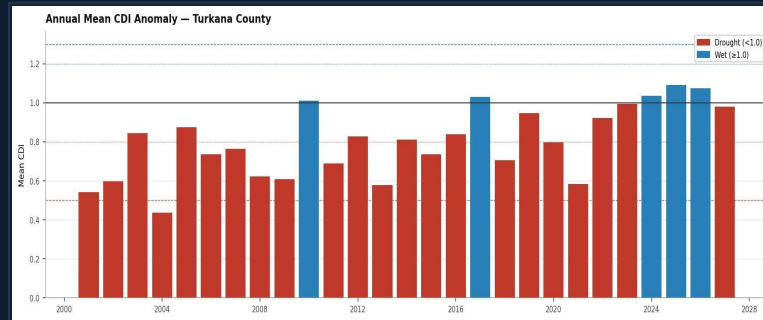
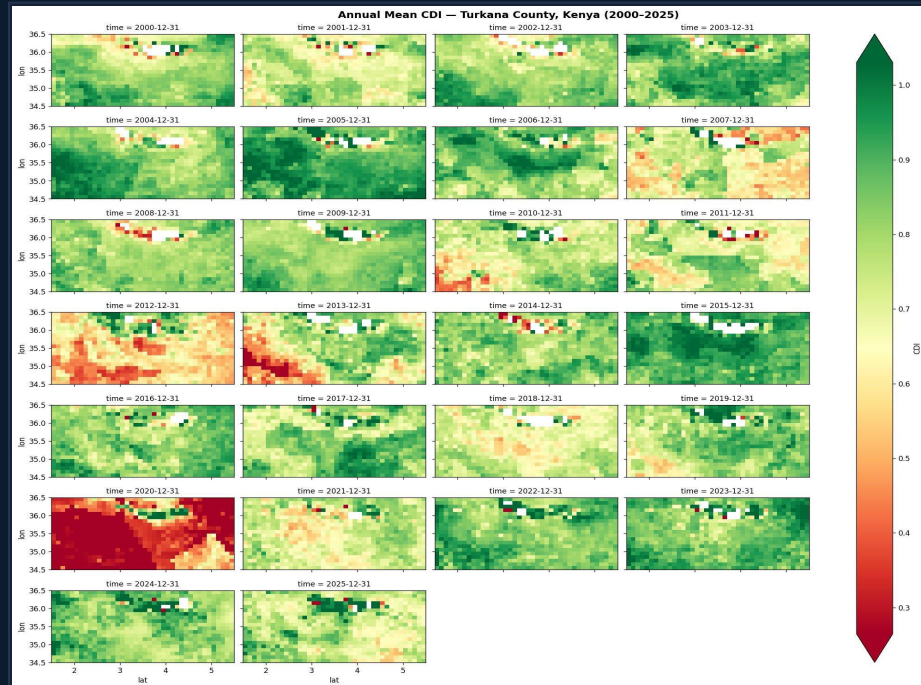
Pixel-wise normalisation and index computation returning annual spatial grids (PDI, TDI, VDI, CDI) per year and month

4 Community Output

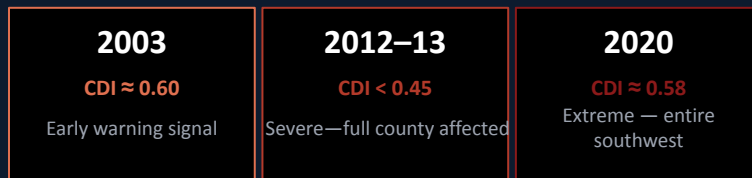
5-class severity map (None → Extreme) + 6-month STL-SARIMA forecast + interactive DSS dashboard

"The same pipeline runs for any country, any country — define the bounding box, run the model, get 25 years of drought intelligence."

25 Years of CDI Maps: Spatial Drought Fingerprints — Turkana County, Kenya

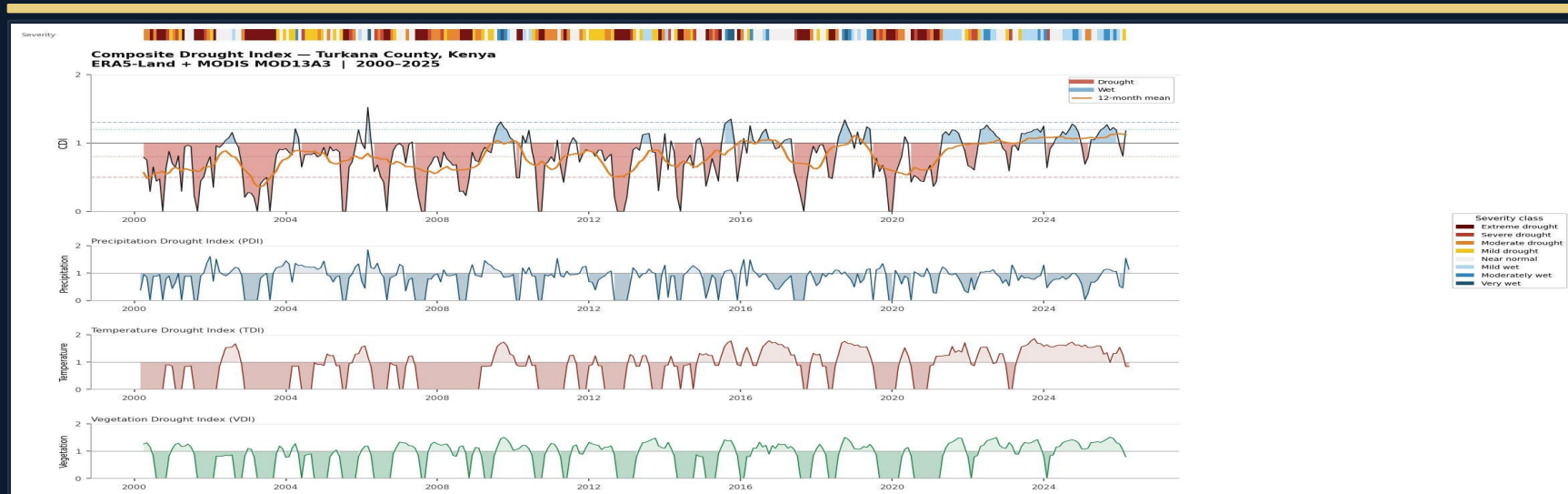


Annual Mean CDI Anomaly · Red = Drought · Blue = Wet Year



- Southwest Turkana is the most drought-vulnerable zone—consistently lowest CDI values
- 2020 La Niña signature: catastrophic extent — CDI collapses across the entire county simultaneously
- Post-2022 recovery visible: 2023–2024 trending green — community adaptation interventions showing results

25-Year Monthly CDI Time-Series: Reading the Drought Rhythm



Composite Drought Index—Monthly CDI with PDI, TDI & VDI Sub-indices (2000–2025) · ERA5-Land + MODIS MOD13A3 · Orange = 12-month rolling mean

PDI

Highly variable — deep troughs in 2004, 2011, 2020–22 align with La Niña events. PDI drives acute drought onset.

TDI

Persistently elevated baseline with a slow rising trend since 2015 — the unmistakable climate change signal.

VDI

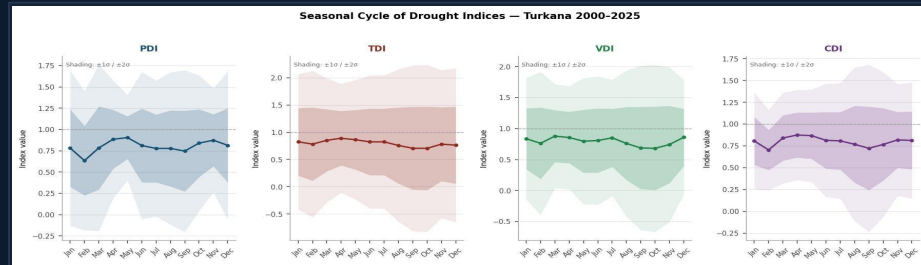
Seasonal oscillations following rainy seasons. Deep troughs in 2009, 2017, and 2020 align directly with crop failures.

CDI

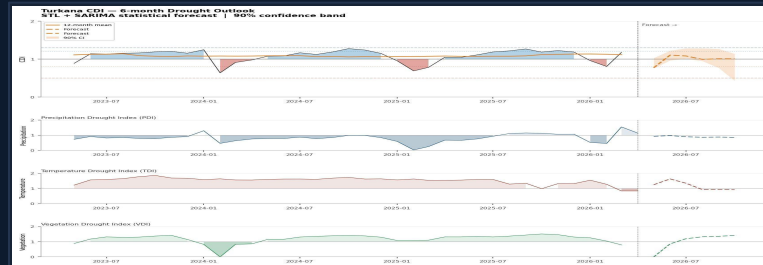
The 12-month mean (orange line) reveals the slow-moving trend. Rising since 2022 — cautious community optimism.

"The severity strip at the top of the chart IS the community early warning: dark red = extreme drought. → immediate response · blue =

Anticipatory Adaptation: Seasonal Patterns, 6-Month Outlook & Scaling Across Africa



Seasonal Cycle of PDI, TDI, VDI & CDI — 25-year climatology (2000–2025)



6-Month Drought Outlook — STL + SARIMA Statistical Forecast · 90% Confidence Band

WHAT SEASONAL PATTERNS TELL COMMUNITIES

- PDI dips in February: planting decisions are most critical in the January–February window
- TDI elevated year-round: Turkana is permanently heat-stressed — drought-tolerant varieties are non-negotiable
- VDI lowest in August–September: pastoral communities must plan water and fodder reserves by June

KEY TAKEAWAYS

Multi-indicator models outperform single-index approaches — drought is compound and multi-dimensional

Temporal depth (25 years) is what separates meaningful anomaly detection from noise

THE REPLICATION BLUEPRINT — FROM TURKANA TO ALL OF AFRICA

1 Any County, Any AOI

Define bounding box for any African region. Run the CDI pipeline—same code, same model, new geography.

2 Build the Baseline

25-year climatology establish seasonal norms. Overlay community context: crop calendar, livelihood zones.

3 Forecast & Translate

STL + SARIMA 6-month outlook. Map CDI to severity classes. Publish via DSS, local radio, or SMS alert system.